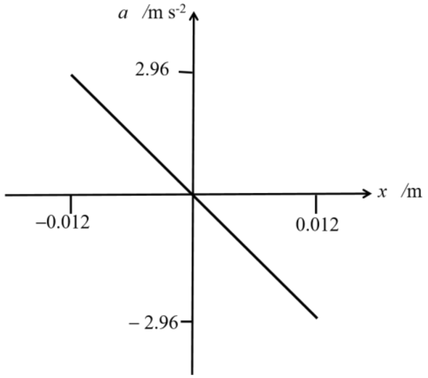
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	*(a)*		a = acceleration ω = angular velocity or angular frequency or pulsatance x = displacement **All 3 correct** (1)	1			1		
	(b)	(i)	$\omega = \frac{2\pi}{0.4}$ (1) [= 15.7 s^{-1}] $a_{max} = \omega^2 A = (15.7)^2 \times 0.012$ (1) = 2.96 [m s^{-2}] (1)	1	1 1		3	2	
		(ii)	 One value of a on graph (1) **ecf** from *(b)*(i) Any straight line (1) Correct position of line (1)		3		3	2	
	(c)		$x = \boxed{0.012} \cos\left(\boxed{15.7}\, t + \boxed{\frac{3\pi}{2}}\right)$ (1 × 3 – one mark for each box) (alternative for angle: $-\frac{\pi}{2}$) Accept 5π for 15.7. **ecf** on ω		3		3	1	
	(d)	(i)	Example e.g. microwave ovens **or** swing (1) Oscillator and driving force named e.g. water molecules and microwaves **or** swing and person pushing (1)	2			2		
		(ii)	Example and consequence e.g. bridge and falling **or** something in the dashboard and buzzing (1) Driving force e.g. wind / soldiers marching **or** engine (1) Resonance explained i.e. both frequencies are the same (1)	3			3		
			Question 3 total	**7**	**8**	**0**	**15**	**5**	**0**